

Business Use Case: AI-Powered Intelligent Ticket Routing using n8n

Project Title

AI-Based Automated Ticket Classification & Department Routing using n8n and LLM

Business Problem

Organizations receive hundreds or thousands of support tickets every day from customers, employees, and business partners. Traditionally, these tickets are manually reviewed by support teams before being assigned to the appropriate department.

This manual process results in:

- Delayed ticket assignment
- Increased operational workload
- Human errors in ticket categorization
- Longer issue resolution times
- Reduced customer satisfaction
- Higher support costs

As ticket volumes increase, manual routing becomes inefficient and difficult to scale.

Objective

Develop an AI-powered workflow that automatically analyzes incoming support tickets, identifies the appropriate business department, and routes the ticket without human intervention.

The primary goals are to:

- Reduce manual ticket triaging
 - Improve routing accuracy
 - Decrease response and resolution time
 - Enhance operational efficiency
 - Scale support operations without increasing manpower
-

Solution

Built an intelligent ticket routing workflow using **n8n** integrated with a **Large Language Model (LLM)**.

The workflow leverages Natural Language Processing (NLP) to understand ticket descriptions and automatically classify them into predefined departments.

Departments include:

- Engineering
- IT & Compliance
- Management
- Cybersecurity
- Operations

Once classified, the workflow routes each ticket to the appropriate department for faster resolution.

Business Workflow

1. A new support ticket is received.
 2. The ticket enters the n8n automation workflow.
 3. The ticket description is sent to the AI model for analysis.
 4. The LLM extracts the ticket intent and predicts the responsible department.
 5. A parser converts the AI response into a structured department label.
 6. The Switch node evaluates the predicted department.
 7. The workflow automatically routes the ticket to:
 - Engineering
 - IT & Compliance
 - Management
 - Cybersecurity
 - Operations
 8. The assigned team receives the ticket immediately for action.
-

Key Features

- AI-powered ticket understanding
 - Automatic department classification
 - No manual ticket triaging
 - Rule-based workflow automation using n8n
 - Intelligent routing using LLM
 - Easy integration with CRM and helpdesk platforms
 - Scalable workflow architecture
 - Low-code automation
-

Business Value Delivered

1. Faster Ticket Assignment

Tickets are routed instantly instead of waiting for manual review, reducing delays in support operations.

2. Reduced Operational Cost

Eliminates repetitive manual work performed by support coordinators, allowing teams to manage higher ticket volumes without increasing staffing.

3. Improved Resolution Time

Correct department assignment on the first attempt enables faster issue resolution and reduces ticket bouncing between teams.

4. Higher Routing Accuracy

The LLM understands natural language, enabling more accurate classification than keyword-based routing systems.

5. Increased Team Productivity

Support teams spend less time sorting tickets and more time resolving customer issues.

6. Better Customer Experience

Customers receive quicker responses because tickets reach the appropriate team immediately.

7. Scalable Support Operations

The workflow can process thousands of tickets automatically without requiring proportional growth in support personnel.

Product Metrics (KPIs)

North Star Metric

Percentage of Tickets Automatically Routed Correctly

Success Metrics

- Ticket Routing Accuracy
- Average First Response Time (FRT)
- Average Resolution Time (ART)

- Manual Routing Reduction (%)
 - Ticket Reassignment Rate
 - SLA Compliance Rate
 - Mean Time to Assign (MTTA)
 - Customer Satisfaction Score (CSAT)
 - Support Team Productivity
 - Monthly Operational Cost Savings
-

Technical Architecture

Automation Platform: n8n

AI Engine: Large Language Model (LLM)

Workflow Components:

- Manual/Webhook Trigger
- AI Ticket Analysis
- Department Parser
- Switch Node for Decision Logic
- Automated Department Routing

Departments Supported:

- Engineering
 - IT & Compliance
 - Management
 - Cybersecurity
 - Operations
-

Expected Business Impact

- 70–90% reduction in manual ticket routing effort
- 40–60% faster ticket assignment
- Improved SLA compliance
- Reduced ticket reassignment between departments
- Higher support team efficiency
- Lower operational costs
- Better customer satisfaction through faster issue resolution